EmpowerHealth Watch Research Field Specification for App + Export

Introduction

We need all research-related app fields stored as structured variables with stable field names and stable export names. Please use coded values for categorical fields and avoid free text except where explicitly noted, such as “Other, specify” fields. All research exports should be linked by study_id, include timestamps where relevant, support CSV and JSON export, and be de-identified (no names or emails). Please also build skip logic for conditional fields, including pregnancy versus postpartum fields and Other-specify fields.

1. Purpose

This document defines the research-related fields that should be built into the EmpowerHealth Watch app, stored in a structured format, included in research exports, and mapped to REDCap where applicable. It is intended to support app development, research data integration, and study operations for the EmpowerHealth Watch pilot.

This document serves three functions. First, it acts as a **field specification** for the developer by listing exact field names, field types, coding, and logic. Second, it acts as a **data dictionary** by defining what each field means and how it should be stored. Third, it acts as a **crosswalk** by showing how app fields align with REDCap variables and study instruments.

This document is based on the current developer research integration updates and the pilot research framework. It should be updated if new research-related app features or export requirements are added.

2. General Data Requirements

All research-linked app data should be exportable by **study ID**. The app should support **de-identified export** and exclude direct identifiers such as participant name and email address from research export files. Export formats should include **CSV** and **JSON**. Timestamps should be included where relevant and stored in a consistent datetime format.

Whenever possible, categorical fields should use fixed coded values rather than open free text. Free text should only be used when a field specifically requires an “Other, specify” response or narrative detail. Skip logic should be built into the app for conditional fields such as pregnancy versus postpartum status, insurance other, or recruitment source other.

The developer update specifies that research export must include user study ID, module completions, needs checklist selections, navigation outcomes, micro ratings, provider review activity, recruitment source, and de-identified CSV/JSON export.

3. Draft Field Table

The table below is the main working field specification for app build, export, and REDCap alignment.

Section	Field name	User-facing label or description	Type	Coding / allowed values	Required	Skip logic / notes	Export name	REDCap variable
User-level	`study_id`	Hidden system field	Text	Unique study ID	Yes	Primary linkage key across app and REDCap	`study_id`	`study_id`
User-level	`research_participant`	Hidden backend toggle	Yes/No	`1=Yes`, `0=No`	Yes	Used for clean research filtering and export	`research_participant`	`research_participant`
User-level	`recruitment_source`	How did you hear about EmpowerHealth Watch?	Single select	`1=Doula`, `2=CHW`, `3=Home Visitor`, `4=CBO`, `5=Social Media`, `6=Event`, `7=Other`	Yes	Onboarding field; store as user-level variable	`recruitment_source`	`recruitment_source`
User-level	`recruitment_source_other`	Other, please specify	Short text	Free text	Conditional	Only if `recruitment_source=7`	`recruitment_source_other`	`recruitment_source_other`
User-level	`recruitment_pathway`	Hidden/admin-assigned pathway field	Single select	`1=Navigator-supported`, `2=Self-directed`	Yes	Framework uses two recruitment pathways for comparison	`recruitment_pathway`	`recruitment_pathway`
Baseline	`age_years`	What is your age?	Integer	Minimum 18	Yes	Recommend exact age in years; can derive age groups later	`age_years`	`age_years`

Baseline	age_group	Derived reporting field	Derived	1=18-25, 2=26-35, 3=36-45, 4=46+	No	Optional derived field for reporting only	age_group	age_group
Baseline	pp_status	Are you currently pregnant or postpartum?	Single select	1=Pregnant, 2=Postpartum	Yes	Baseline research field	pp_status	pp_status
Baseline	gest_week	How many weeks pregnant are you?	Integer	Suggested range 1-45	Conditional	Only if pp_status=1	gest_week	gest_week
Baseline	postpartum_month	How many months postpartum are you?	Integer	Suggested range 0-12	Conditional	Only if pp_status=2	postpartum_month	postpartum_month
Baseline	insurance_type	What type of insurance do you currently have?	Single select	1=Medicaid, 2=Private, 3=Uninsured, 4=Other	Yes	Baseline research field	insurance_type	insurance_type
Baseline	insurance_other	Other insurance, please specify	Short text	Free text	Conditional	Only if insurance_type=4	insurance_other	insurance_other
Baseline	support_person_nav	Do you have a support person who helps you with healthcare or care navigation?	Yes/No	1=Yes, 0=No	Yes	Baseline research field	support_person_nav	support_person_nav
Baseline	baseline_advocacy_conf	How confident do you feel speaking up in healthcare visits?	Likert	1=Not at all confident, 2=A little confident, 3=Somewhat confident, 4=Very confident, 5=Extremely confident	Yes if collected in app	Could remain REDCap-only if preferred	baseline_advocacy_conf	baseline_advocacy_conf

Micro measures	`micro_understand`	I understand what this means.	Likert 1–5	`1=Strongly disagree`, `2=Disagree`, `3=Neutral`, `4=Agree`, `5=Strongly agree`	Yes when shown	Collected after modules and/or AVS	`micro_understand`	`micro_understand`
Micro measures	`micro_next_step`	I know what I need to do next.	Likert 1–5	`1=Strongly disagree`, `2=Disagree`, `3=Neutral`, `4=Agree`, `5=Strongly agree`	Yes when shown	Collected after modules and/or AVS	`micro_next_step`	`micro_next_step`
Micro measures	`micro_confidence`	I feel confident about my next steps.	Likert 1–5	`1=Strongly disagree`, `2=Disagree`, `3=Neutral`, `4=Agree`, `5=Strongly agree`	Yes when shown	Collected after modules and/or AVS	`micro_confidence`	`micro_confidence`
Micro measures	`content_id`	Hidden content identifier	Text	Module ID or visit summary ID	Yes	Links rating event to content item	`content_id`	`content_id`
Micro measures	`content_type`	Hidden content type	Single select	`1=Module`, `2=Visit summary`	Yes	Required for event-level export	`content_type`	`content_type`
Micro measures	`micro_ts`	Hidden timestamp	Datetime	ISO datetime	Yes	Timestamp for each micro rating event	`micro_ts`	`micro_ts`
Needs checklist	`need_prenatal_postpartum`	Need help with prenatal/post partum care	Checkbox/ binary	`1=Selected`, `0=Not selected`	No	One field per need category	`need_prenatal_postpartum`	`need_prenatal_postpartum`
Needs checklist	`need_delivery_prep`	Need help with delivery preparation	Checkbox/ binary	`1=Selected`, `0=Not selected`	No	One field per need category	`need_delivery_prep`	`need_delivery_prep`

Needs checklist	need_med_followup	Need help with medical condition follow-up	Checkbox/ binary	1=Selected, 0=Not selected	No	One field per need category	need_med_followup	need_med_followup
Needs checklist	need_mental_health	Need help with mental health support	Checkbox/ binary	1=Selected, 0=Not selected	No	One field per need category	need_mental_health	need_mental_health
Needs checklist	need_lactation	Need help with lactation/feeding support	Checkbox/ binary	1=Selected, 0=Not selected	No	One field per need category	need_lactation	need_lactation
Needs checklist	need_infant_care	Need help with infant care	Checkbox/ binary	1=Selected, 0=Not selected	No	One field per need category	need_infant_care	need_infant_care
Needs checklist	need_benefits	Need help with benefits/resources	Checkbox/ binary	1=Selected, 0=Not selected	No	One field per need category	need_benefits	need_benefits
Needs checklist	need_transport	Need help with transportation/logistics	Checkbox/ binary	1=Selected, 0=Not selected	No	One field per need category	need_transport	need_transport
Needs checklist	need_other	Other need	Checkbox/ binary	1=Selected, 0=Not selected	No	If selected, show specify field	need_other	need_other
Needs checklist	need_other_text	Other, please specify	Short text	Free text	Conditional	Only if need_other=1	need_other_text	need_other_text
Navigation outcome	need_prenatal_postpartum_outcome	Did you get what you needed?	Single select	1=Yes, 2=Partly, 3=No, 4=Didnt_try, 5=Didnt_know_how, 6=Couldnt_access	Conditional	Only if need selected	need_prenatal_postpartum_outcome	need_prenatal_postpartum_outcome
Navigation outcome	need_delivery_prep_outcome	Did you get what you needed?	Single select	Same coding as above	Conditional	Only if need selected	need_delivery_prep_outcome	need_delivery_prep_outcome

Navigation outcome	need_med_followup_outcome	Did you get what you needed?	Single select	Same coding as above	Conditional	Only if need selected	need_med_followup_outcome	need_med_followup_outcome
Navigation outcome	need_mental_health_outcome	Did you get what you needed?	Single select	Same coding as above	Conditional	Only if need selected	need_mental_health_outcome	need_mental_health_outcome
Navigation outcome	need_lactation_outcome	Did you get what you needed?	Single select	Same coding as above	Conditional	Only if need selected	need_lactation_outcome	need_lactation_outcome
Navigation outcome	need_infant_care_outcome	Did you get what you needed?	Single select	Same coding as above	Conditional	Only if need selected	need_infant_care_outcome	need_infant_care_outcome
Navigation outcome	need_benefits_outcome	Did you get what you needed?	Single select	Same coding as above	Conditional	Only if need selected	need_benefits_outcome	need_benefits_outcome
Navigation outcome	need_transport_outcome	Did you get what you needed?	Single select	Same coding as above	Conditional	Only if need selected	need_transport_outcome	need_transport_outcome
Navigation outcome	need_other_outcome	Did you get what you needed?	Single select	Same coding as above	Conditional	Only if need selected	need_other_outcome	need_other_outcome
Needs checklist	needs_ts	Hidden timestamp	Datetime	ISO datetime	Yes	Timestamp for needs checklist event	needs_ts	needs_ts
Milestone prompts	milestone_health_question	Did you have a health question recently?	Yes/No	1=Yes, 0=No	Yes when shown	Schedulable prompt	milestone_health_question	milestone_health_question
Milestone prompts	milestone_clear_next_step	Did you feel clear on what to do next?	Yes/No	1=Yes, 0=No	Yes when shown	Schedulable prompt	milestone_clear_next_step	milestone_clear_next_step
Milestone prompts	milestone_app_he	Did the app help you take your next step?	Yes/No	1=Yes, 0=No	Yes when shown	Schedulable prompt	milestone_app_help	milestone_app_helped_next_step

	lped_next_step						ed_next_step	
Milestone prompts	milestone_type	Hidden milestone context	Single select	1=Late pregnancy, 2=Postpartum, 3=Other	Yes	Used for analysis and scheduling context	milestone_type	milestone_type
Milestone prompts	milestone_ts	Hidden timestamp	Datetime	ISO datetime	Yes	Timestamp for milestone event	milestone_ts	milestone_ts
App activity	module_completion	Hidden event log	Repeating event/log	Module completion record	Yes for export	Export should include module ID and completion timestamp	module_completion	module_completion
App activity	provider_review_activity	Hidden event log	Repeating event/log	Provider review record	Yes for export	Required in developer export list	provider_review_activity	provider_review_activity
AVS upload	avs_upload_type	Hidden event attribute	Single select	1=PDF upload, 2=Camera photo	If AVS uploaded	Reflects request for both upload and photo capture	avs_upload_type	avs_upload_type
Plain-language library	health_made_simple_access	Hidden event log	Repeating event/log	Search/access record				

4. REDCap Mapping Notes

REDCap should remain the main home for study consent and survey-based research instruments. The app should hold research-linked operational and event-level data that can either be imported into REDCap or maintained as a linked analytic dataset using `study_id`.

Recommended REDCap-side structure:

Baseline instrument

- `study_id`
- demographics
- `pp_status`
- `gest_week`
- `postpartum_month`
- `insurance_type`
- `support_person_nav`
- baseline self-advocacy confidence item(s)
- recruitment source
- recruitment pathway

Endline instrument

- endline AURA
- self-advocacy items
- care navigation outcomes
- safety items
- whether anyone helped participant use the app
- who helped

Milestone check-in instrument

- `milestone_health_question`
- `milestone_clear_next_step`
- `milestone_app_helped_next_step`

Repeating app data instrument or linked import file

- micro ratings
- needs checklist selections
- navigation outcomes
- module completion events
- provider review activity
- timestamps

This structure aligns with the research framework's baseline, milestone, and endline data collection plan while also reflecting the app export requirements described in the developer update.

5. Developer Quick-Reference Coding List

This section is a shorthand summary for the developer. It does not replace the full field table above. It is intended as a quick-reference list of the main coded fields.

study_id
text

research_participant
radio
1, Yes | 0, No

recruitment_source
radio
1, Doula | 2, CHW | 3, Home Visitor | 4, CBO | 5, Social Media | 6, Event | 7, Other

recruitment_source_other
text

recruitment_pathway
radio
1, Navigator-supported | 2, Self-directed

age_years
text (integer, min 18)

age_group
radio
1, 18-25 | 2, 26-35 | 3, 36-45 | 4, 46+

pp_status
radio
1, Pregnant | 2, Postpartum

gest_week
text (integer)

postpartum_month
text (integer)

insurance_type
radio
1, Medicaid | 2, Private | 3, Uninsured | 4, Other

insurance_other
text

support_person_nav
radio
1, Yes | 0, No

baseline_advocacy_conf
radio
1, Not at all confident | 2, A little confident | 3, Somewhat confident | 4, Very confident | 5, Extremely confident

micro_understand

radio
1, Strongly disagree | 2, Disagree | 3, Neutral | 4, Agree | 5, Strongly agree

micro_next_step
radio
1, Strongly disagree | 2, Disagree | 3, Neutral | 4, Agree | 5, Strongly agree

micro_confidence
radio
1, Strongly disagree | 2, Disagree | 3, Neutral | 4, Agree | 5, Strongly agree

6. Closing Note

This document is the working specification for research-linked app fields, export requirements, and REDCap alignment for the EmpowerHealth Watch pilot. It should be updated as the app evolves and as the study team finalizes any remaining baseline demographic decisions, derived reporting categories, or REDCap import workflows.

The REDCap instrument/data dictionary should be maintained as a separate companion document using these same field names, coding schemes, and mappings.